



Executive Summary

This project analyzes a Retail Sales Dataset to uncover customer purchasing behavior, sales performance trends, and product category insights. The dataset was cleaned, explored, and visualized to transform raw transactional data into meaningful business insights that can support data-driven decision-making.

Through exploratory data analysis, several important patterns were identified regarding customer demographics, sales performance, and product demand. The analysis revealed how different customer segments contribute to revenue, which product categories perform best, and how purchasing behavior changes over time.



Key Insights



Sales Trend Analysis

- Sales showed noticeable variations across different months and periods, indicating seasonal purchasing patterns and changing customer demand.



Product Category Performance

- Clothing emerged as one of the top-performing product categories, generating a significant share of overall sales and customer purchases.



Customer Demographics

- Adult customers represented the largest customer segment and contributed the highest portion of sales.
- Purchasing behavior differed across age groups, highlighting the importance of customer segmentation.



Gender-Based Analysis

- Both male and female customers contributed significantly to total sales, with observable differences in purchasing preferences and spending behavior.



Top Customer Identification

- A small group of high-value customers generated a substantial portion of total revenue, emphasizing the importance of customer retention strategies.



Quantity vs Revenue Relationship

- A strong positive relationship was observed between product quantity purchased and total sales amount, indicating that larger purchase volumes directly contribute to higher revenue.



Customer Purchasing Patterns

- Sales activity varied across different days and months, providing insights into peak shopping periods and customer buying habits.

Business Value

- The findings can help businesses optimize inventory planning, improve marketing campaigns, identify target customer groups, and make informed sales decisions.

Conclusion

The analysis successfully transformed raw retail transaction data into actionable business insights. By identifying sales trends, customer behavior patterns, top-performing product categories, and high-value customers, this project demonstrates how data analytics can support strategic business decisions and drive revenue growth. The visualizations further enhanced the understanding of key trends, making complex data easier to interpret and communicate.